

Seminar 1[†]

Describing Data 1 (Data Visualization)

Gabriel E. Cabrera Guzmán 

The University of Manchester
Alliance Manchester Business School

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[†]gabriel.cabreraguzman@postgrad.manchester.ac.uk

Exercise 1

What sort of display would be most suitable for the following data?

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2. The **types** of premises occupied by all the small businesses in a town – **office, factory, warehouse**, etc.

A: These are categorical variables. Hence, Bar Chart or Pie Chart would be the most suitable visualisation.

Exercise 1 (Cont'd)

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4. The **number of trainee accountants** enrolling for professional examinations **each year since 1960**.

A: Although there is only one numerical measure, but it must be illustrated with respect to time. Therefore, Time Series Plot would be most suitable.

Exercise 1 (Cont'd)

5. The **number of years of experience** and **the current salary** of a random sample of economists working for a New York bank.

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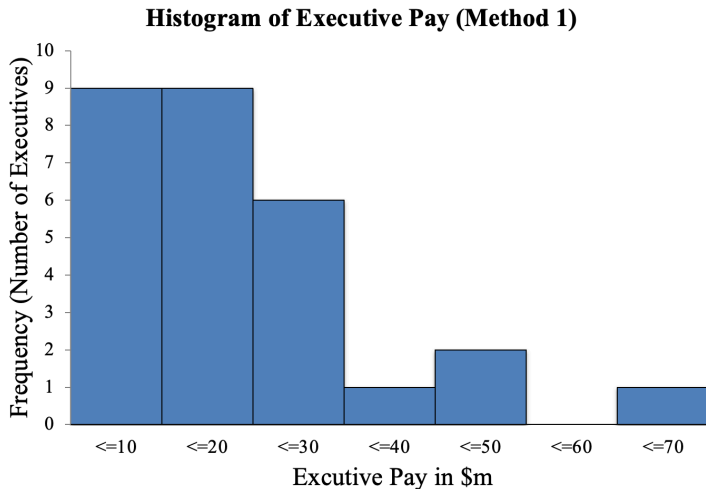
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Exercise 2

In one year, the executive pay, including salary and bonus, in millions of dollars, for 28 chairpersons and chief executive officers are presented in the table below. Display these data by drawing a histogram using Excel.

Company	Value	Company	Value
Boeing	21.12	Coca-Cola	21.60
Whirlpool	12.93	DuPont	12.73
Bank of America	12.00	Motorola	47.15
Sherwin-Williams	11.00	Marriott	8.58
Bristol-Myers Squibb	16.00	Honeywell	16.28
General Mills	12.19	Exxon	34.90
Hillshire Brands	6.40	CBS	60.25
Wal-Mart	20.70	AT&T	21.00
Apple	4.17	Philip Morris	24.73
Goodyear	13.43	Tesco	1.90
Teledyne	3.21	EasyJet	6.40
Delta Airlines	12.58	Shell	5.10
Chrysler	1.23	GlaxoSmithKline	6.39

Exercise 2 (Cont'd)



Exercise 3

A financial analyst is interested in the amount of resources spent by computer companies on Research and Development (R&D). She samples 30 such firms and calculates the percentage of total revenue they spent on R&D in the last year. The results are given below. Summarize these data using by using: a) the frequencies, b) the relative frequencies, c) a histogram.

Company	(%)	Company	(%)	Company	(%)
1	6.0	11	7.9	21	8.0
2	10.4	12	6.8	22	7.7
3	10.5	13	7.4	23	7.4
4	9.0	14	9.5	24	6.5
5	7.3	15	8.1	25	9.5
6	6.6	16	13.5	26	8.2
7	6.9	17	9.9	27	6.9
8	8.2	18	6.9	28	7.2
9	7.1	19	11.1	29	8.2
10	8.1	20	8.2	30	6.7

Also, use the histogram to deduce the proportion of companies that spend **more than 9%** on R&D.

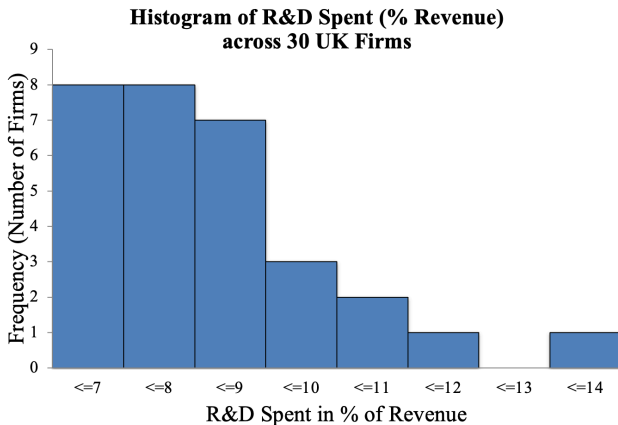
Exercise 3 (Cont'd)

A: For a) and b)

Bin	Frequency	Relative Frequency
7	8	26.7%
8	8	26.7%
9	7	23.3%
10	3	10.0%
11	2	6.7%
12	1	3.3%
13	0	0.0%
14	1	3.3%
Total	30	100%

Exercise 3 (Cont'd)

For c)



Therefore, 7/30 companies spend 9% or more on R&D.

Exercise 4

Choosing a suitable leaf unit, draw a stem and leaf diagram **by hand** for the list of executive salaries given below in thousands of pounds. The data are:

846	457	563	1466	1200
2164	746	1611	824	824
1310	1007	1367	575	1252
1354	2479	1238	927	1253
925	1284	1279	1660	860

Exercise 4 (Cont'd)

A: Taking a stem unit of 1000 and leaf unit of 100 gives only three classes but allowing two rows for each stem unit overcomes this. We have rounded to obtain the leaf unit. For example, the first data point (846) can be approximated to 800, which is written in the form of $0 \times 1000 + 8 \times 100$. This means that the number of stem is zero and there are 8 leaves. Consider another example: 2164 is approximated to 2200 and is written as $2 \times 1000 + 2 \times 100$. Repeating this, the original data becomes:

800	500	600	1500	1200
2200	700	1600	800	800
1300	1000	1400	600	1300
1400	2500	1200	900	1300
900	1300	1300	1700	900

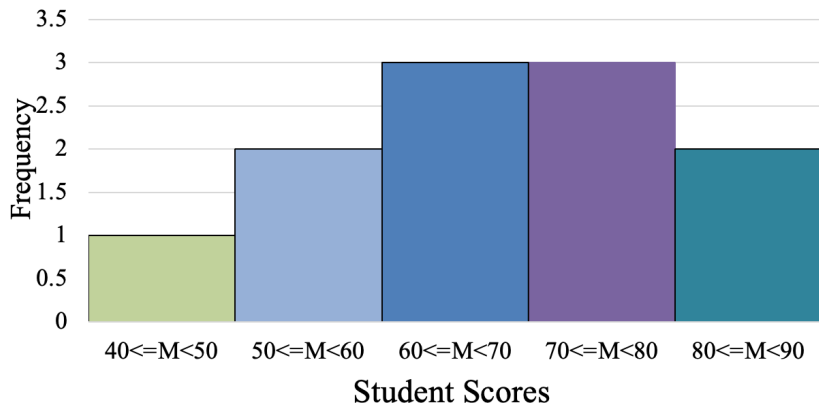
Exercise 4 (Cont'd)

A: The above data can be presented as follows:

Stem ($\times 1000$)	Leaves ($\times 100$)
0	8 5 6 7 8 8 6 9 9 9
1	2 3 0 4 3 4 2 3 3 3
1	5 6 7
2	2
2	5

Exercise 5

Estimate the average, median, and standard deviation of the original data, which is presented by the following histogram.



Exercise 5 (Cont'd)

A: We can only guess what the data points in each bin are. For instance, a data point located in the first bin ($40 \leq M < 50$) can be 40, 40.12, 45, etc. A reasonable way is to assume that this data point is in the middle of the first bin. Therefore, we assume the first data point is 45. Using the same argument, we assume that the two data points located in the second bin are equal to the middle of the second bin (55). Thus, the data set can be assumed as follows:

1	2	3	4	5	6	7	8	9	10	11
45	55	55	65	65	65	75	75	75	85	85

Hence,

- Mean = $\frac{\sum_{i=1}^n X_i}{n} = \frac{45+55+\dots+85+85}{11} = \frac{745}{11} = 67.727$
- Median = $X_{\frac{11+1}{2}} = X_6 = 65$
- Mode: 65 and 75

Exercise 5 (Cont'd)

- Variance:

$$\begin{aligned} S^2 &= \frac{\sum_{i=1}^n (X_i - \bar{X})^2}{n - 1} \\ &= \frac{(45 - 67.72)^2 + (55 - 67.72)^2 + \dots + (85 - 67.72)^2}{11 - 1} \\ &= \frac{1618.18}{10} = 161.81 \end{aligned}$$

- Standard Deviation: $S = \sqrt{161.81} = 12.720$